SQL Handbook (351–400 Commands with Before & After Tables - Exact Model)

351 ADD COLUMN AFTER

```
Before Table (students):
| id | name | age | city |
| 1 | Ravi | 20 | Chennai |
| 2 | Kumar | 21 | Madurai |
| 3 | Anu | 19 | Coimbatore |
```

```
SQL:
ALTER TABLE students ADD COLUMN email VARCHAR(100) AFTER name;
```

```
After:
Column added after name
```

352 ADD COLUMN FIRST

```
Before Table (students):
| id | name | age | city |
| 1 | Ravi | 20 | Chennai |
| 2 | Kumar | 21 | Madurai |
| 3 | Anu | 19 | Coimbatore |
```

```
SQL:
ALTER TABLE students ADD COLUMN code INT FIRST;
```

```
After:
Column added first
```

353 MODIFY NULL

```
Before Table (students):
| id | name | age | city |
| 1 | Ravi | 20 | Chennai |
| 2 | Kumar | 21 | Madurai |
| 3 | Anu | 19 | Coimbatore |
```

```
SQL:
ALTER TABLE students MODIFY city VARCHAR(50) NULL;
```

```
After:
Column allows NULL
```

354 MODIFY NOT NULL

```
Before Table (students):
| id | name | age | city |
| 1 | Ravi | 20 | Chennai |
| 2 | Kumar | 21 | Madurai |
| 3 | Anu | 19 | Coimbatore |
```

```
SQL:
ALTER TABLE students MODIFY city VARCHAR(50) NOT NULL;
```

```
After:
Column not null
```

355 CHANGE COLUMN

```
Before Table (students):
```

```
| id | name | age | city |
| 1 | Ravi | 20 | Chennai |
| 2 | Kumar | 21 | Madurai |
| 3 | Anu | 19 | Coimbatore |
```

SQL:

```
ALTER TABLE students CHANGE name full_name VARCHAR(100);
```

After:
Column renamed & type changed

356 RENAME TABLE ALT

Before Table (students):

```
| id | name | age | city |
| 1 | Ravi | 20 | Chennai |
| 2 | Kumar | 21 | Madurai |
| 3 | Anu | 19 | Coimbatore |
```

SQL:

```
ALTER TABLE students RENAME students_old;
```

After:
Table renamed

357 AUTO_INCREMENT SET

Before Table (students):

```
| id | name | age | city |
| 1 | Ravi | 20 | Chennai |
| 2 | Kumar | 21 | Madurai |
| 3 | Anu | 19 | Coimbatore |
```

SQL:

```
ALTER TABLE students AUTO_INCREMENT=100;
```

After:
Next id set to 100

358 ADD CHECK

Before Table (students):

```
| id | name | age | city |
| 1 | Ravi | 20 | Chennai |
| 2 | Kumar | 21 | Madurai |
| 3 | Anu | 19 | Coimbatore |
```

SQL:

```
ALTER TABLE students ADD CHECK (age BETWEEN 0 AND 120);
```

After:
Check added

359 DROP CHECK

Before Table (students):

```
| id | name | age | city |
| 1 | Ravi | 20 | Chennai |
| 2 | Kumar | 21 | Madurai |
| 3 | Anu | 19 | Coimbatore |
```

SQL:

```
ALTER TABLE students DROP CHECK age;
```

After:
Check removed

360 ADD FK CASCADE

```
Before Table (students):
| id | name | age | city |
| 1 | Ravi | 20 | Chennai |
| 2 | Kumar | 21 | Madurai |
| 3 | Anu | 19 | Coimbatore |
SQL:
ALTER TABLE students ADD CONSTRAINT fk_dept FOREIGN KEY(id) REFERENCES dept(id) ON DELETE CASCADE;
After:
FK cascade added
```

361 ADD FK SET NULL

```
Before Table (students):
| id | name | age | city |
| 1 | Ravi | 20 | Chennai |
| 2 | Kumar | 21 | Madurai |
| 3 | Anu | 19 | Coimbatore |
SQL:
ALTER TABLE students ADD CONSTRAINT fk_dept2 FOREIGN KEY(id) REFERENCES dept(id) ON DELETE SET NULL;
After:
FK set null
```

362 SHOW CREATE DB

```
Before Table (students):
| id | name | age | city |
| 1 | Ravi | 20 | Chennai |
| 2 | Kumar | 21 | Madurai |
| 3 | Anu | 19 | Coimbatore |
SQL:
SHOW CREATE DATABASE test;
After:
DDL of database
```

363 USE DB

```
Before Table (students):
| id | name | age | city |
| 1 | Ravi | 20 | Chennai |
| 2 | Kumar | 21 | Madurai |
| 3 | Anu | 19 | Coimbatore |
SQL:
USE test;
After:
Database selected
```

364 SHOW TABLES LIKE

```
Before Table (students):
| id | name | age | city |
| 1 | Ravi | 20 | Chennai |
| 2 | Kumar | 21 | Madurai |
| 3 | Anu | 19 | Coimbatore |
SQL:
SHOW TABLES LIKE 'stud%';
After:
```

Filtered tables

365 SHOW COLUMNS LIKE

Before Table (students):
id	name	age	city
1	Ravi	20	Chennai
2	Kumar	21	Madurai
3	Anu	19	Coimbatore

SQL:
```
SHOW COLUMNS FROM students LIKE 'n%';
```

After:
Filtered columns

366 DESCRIBE EXTENDED

Before Table (students):
id	name	age	city
1	Ravi	20	Chennai
2	Kumar	21	Madurai
3	Anu	19	Coimbatore

SQL:
```
DESCRIBE EXTENDED students;
```

After:
Extended description

367 EXPLAIN FORMAT

Before Table (students):
id	name	age	city
1	Ravi	20	Chennai
2	Kumar	21	Madurai
3	Anu	19	Coimbatore

SQL:
```
EXPLAIN FORMAT=JSON SELECT * FROM students;
```

After:
JSON plan

368 EXPLAIN PARTITIONS

Before Table (students):
id	name	age	city
1	Ravi	20	Chennai
2	Kumar	21	Madurai
3	Anu	19	Coimbatore

SQL:
```
EXPLAIN PARTITIONS SELECT * FROM students;
```

After:
Partitions info

369 OPTIMIZER TRACE

Before Table (students):
id	name	age	city
1	Ravi	20	Chennai
2	Kumar	21	Madurai
3	Anu	19	Coimbatore

```
SQL:
SET optimizer_trace='enabled=on';
After:
Trace enabled
```

370 SHOW OPTIMIZER TRACE

```
Before Table (students):
| id | name | age | city |
| 1 | Ravi | 20 | Chennai |
| 2 | Kumar | 21 | Madurai |
| 3 | Anu | 19 | Coimbatore |
SQL:
SELECT * FROM information_schema.OPTIMIZER_TRACE;
After:
Trace output
```

371 SET AUTOCOMMIT

```
Before Table (students):
| id | name | age | city |
| 1 | Ravi | 20 | Chennai |
| 2 | Kumar | 21 | Madurai |
| 3 | Anu | 19 | Coimbatore |
SQL:
SET autocommit=0;
After:
Autocommit off
```

372 START TRANSACTION RW

```
Before Table (students):
| id | name | age | city |
| 1 | Ravi | 20 | Chennai |
| 2 | Kumar | 21 | Madurai |
| 3 | Anu | 19 | Coimbatore |
SQL:
START TRANSACTION READ WRITE;
After:
Txn started
```

373 SET ISOLATION

```
Before Table (students):
| id | name | age | city |
| 1 | Ravi | 20 | Chennai |
| 2 | Kumar | 21 | Madurai |
| 3 | Anu | 19 | Coimbatore |
SQL:
SET SESSION TRANSACTION ISOLATION LEVEL READ COMMITTED;
After:
Isolation set
```

374 SHOW VARIABLES AUTOCOMMIT

```
Before Table (students):
| id | name | age | city |
```

```
| 1 | Ravi | 20 | Chennai |
| 2 | Kumar | 21 | Madurai |
| 3 | Anu | 19 | Coimbatore |
```

SQL:

```
SHOW VARIABLES LIKE 'autocommit';
```

After:
Autocommit value

375 LOCK TABLE WRITE

Before Table (students):

```
| id | name | age | city |
| 1 | Ravi | 20 | Chennai |
| 2 | Kumar | 21 | Madurai |
| 3 | Anu | 19 | Coimbatore |
```

SQL:

```
LOCK TABLES students WRITE;
```

After:
Write lock

376 UNLOCK

Before Table (students):

```
| id | name | age | city |
| 1 | Ravi | 20 | Chennai |
| 2 | Kumar | 21 | Madurai |
| 3 | Anu | 19 | Coimbatore |
```

SQL:

```
UNLOCK TABLES;
```

After:
Unlocked

377 CREATE TEMP TABLE

Before Table (students):

```
| id | name | age | city |
| 1 | Ravi | 20 | Chennai |
| 2 | Kumar | 21 | Madurai |
| 3 | Anu | 19 | Coimbatore |
```

SQL:

```
CREATE TEMPORARY TABLE tmp AS SELECT * FROM students;
```

After:
Temp table created

378 DROP TEMP TABLE

Before Table (students):

```
| id | name | age | city |
| 1 | Ravi | 20 | Chennai |
| 2 | Kumar | 21 | Madurai |
| 3 | Anu | 19 | Coimbatore |
```

SQL:

```
DROP TEMPORARY TABLE tmp;
```

After:
Temp table dropped

379 INSERT SELECT

```
Before Table (students):
| id | name | age | city |
| 1 | Ravi | 20 | Chennai |
| 2 | Kumar | 21 | Madurai |
| 3 | Anu | 19 | Coimbatore |

SQL:
INSERT INTO students(id,name,age,city) SELECT id,name,age,city FROM students;

After:
Rows duplicated
```

380 REPLACE INTO

```
Before Table (students):
| id | name | age | city |
| 1 | Ravi | 20 | Chennai |
| 2 | Kumar | 21 | Madurai |
| 3 | Anu | 19 | Coimbatore |

SQL:
REPLACE INTO students VALUES (1,'Ravi',22,'Chennai');

After:
Upsert done
```

381 INSERT IGNORE

```
Before Table (students):
| id | name | age | city |
| 1 | Ravi | 20 | Chennai |
| 2 | Kumar | 21 | Madurai |
| 3 | Anu | 19 | Coimbatore |

SQL:
INSERT IGNORE INTO students VALUES (1,'Ravi',20,'Chennai');

After:
Ignored duplicate
```

382 ON DUP KEY UPDATE

```
Before Table (students):
| id | name | age | city |
| 1 | Ravi | 20 | Chennai |
| 2 | Kumar | 21 | Madurai |
| 3 | Anu | 19 | Coimbatore |

SQL:
INSERT INTO students VALUES (1,'Ravi',20,'Chennai') ON DUPLICATE KEY UPDATE age=age+1;

After:
Updated on dup
```

383 DELETE JOIN

```
Before Table (students):
| id | name | age | city |
| 1 | Ravi | 20 | Chennai |
| 2 | Kumar | 21 | Madurai |
| 3 | Anu | 19 | Coimbatore |

SQL:
DELETE s FROM students s JOIN dept d ON s.id=d.id;

After:
```

```
Deleted via join
```

384 UPDATE JOIN

```
Before Table (students):
| id | name | age | city |
| 1 | Ravi | 20 | Chennai |
| 2 | Kumar | 21 | Madurai |
| 3 | Anu | 19 | Coimbatore |
SQL:
UPDATE students s JOIN dept d ON s.id=d.id SET s.age=s.age+1;
After:
Updated via join
```

385 MERGE (simulate)

```
Before Table (students):
| id | name | age | city |
| 1 | Ravi | 20 | Chennai |
| 2 | Kumar | 21 | Madurai |
| 3 | Anu | 19 | Coimbatore |
SQL:
INSERT INTO students SELECT * FROM staging ON DUPLICATE KEY UPDATE age=VALUES(age);
After:
Merge simulated
```

386 CREATE INDEX BTREE

```
Before Table (students):
| id | name | age | city |
| 1 | Ravi | 20 | Chennai |
| 2 | Kumar | 21 | Madurai |
| 3 | Anu | 19 | Coimbatore |
SQL:
CREATE INDEX idx_age ON students(age) USING BTREE;
After:
Index created
```

387 CREATE INDEX HASH

```
Before Table (students):
| id | name | age | city |
| 1 | Ravi | 20 | Chennai |
| 2 | Kumar | 21 | Madurai |
| 3 | Anu | 19 | Coimbatore |
SQL:
CREATE INDEX idx_age_hash ON students(age) USING HASH;
After:
Hash index
```

388 DROP INDEX ALT

```
Before Table (students):
| id | name | age | city |
| 1 | Ravi | 20 | Chennai |
| 2 | Kumar | 21 | Madurai |
| 3 | Anu | 19 | Coimbatore |
```

```
SQL:
ALTER TABLE students DROP INDEX idx_age;
```

```
After:
Index dropped
```

389 ANALYZE HISTOGRAM

```
Before Table (students):
| id | name | age | city |
| 1 | Ravi | 20 | Chennai |
| 2 | Kumar | 21 | Madurai |
| 3 | Anu | 19 | Coimbatore |
```

```
SQL:
ANALYZE TABLE students UPDATE HISTOGRAM ON age WITH 10 BUCKETS;
```

```
After:
Histogram updated
```

390 SHOW HISTOGRAM

```
Before Table (students):
| id | name | age | city |
| 1 | Ravi | 20 | Chennai |
| 2 | Kumar | 21 | Madurai |
| 3 | Anu | 19 | Coimbatore |
```

```
SQL:
SELECT * FROM information_schema.COLUMN_STATISTICS;
```

```
After:
Histogram view
```

391 SET SQL_SAFE_UPDATES

```
Before Table (students):
| id | name | age | city |
| 1 | Ravi | 20 | Chennai |
| 2 | Kumar | 21 | Madurai |
| 3 | Anu | 19 | Coimbatore |
```

```
SQL:
SET SQL_SAFE_UPDATES=1;
```

```
After:
Safe updates on
```

392 SHOW WARNINGS COUNT

```
Before Table (students):
| id | name | age | city |
| 1 | Ravi | 20 | Chennai |
| 2 | Kumar | 21 | Madurai |
| 3 | Anu | 19 | Coimbatore |
```

```
SQL:
SHOW COUNT(*) WARNINGS;
```

```
After:
Warnings count
```

393 SHOW ERRORS COUNT

```
Before Table (students):
| id | name | age | city |
```

```
| 1 | Ravi | 20 | Chennai |
| 2 | Kumar | 21 | Madurai |
| 3 | Anu | 19 | Coimbatore |
```

```
SQL:
SHOW COUNT(*) ERRORS;
```

```
After:
Errors count
```

394 RESET PERSIST

```
Before Table (students):
| id | name | age | city |
| 1 | Ravi | 20 | Chennai |
| 2 | Kumar | 21 | Madurai |
| 3 | Anu | 19 | Coimbatore |
```

```
SQL:
RESET PERSIST;
```

```
After:
Persist reset
```

395 SET PERSIST

```
Before Table (students):
| id | name | age | city |
| 1 | Ravi | 20 | Chennai |
| 2 | Kumar | 21 | Madurai |
| 3 | Anu | 19 | Coimbatore |
```

```
SQL:
SET PERSIST max_connections=200;
```

```
After:
Persist var set
```

396 SHOW VARIABLES GLOBAL

```
Before Table (students):
| id | name | age | city |
| 1 | Ravi | 20 | Chennai |
| 2 | Kumar | 21 | Madurai |
| 3 | Anu | 19 | Coimbatore |
```

```
SQL:
SHOW GLOBAL VARIABLES LIKE 'max_connections';
```

```
After:
Global var
```

397 SHOW STATUS GLOBAL

```
Before Table (students):
| id | name | age | city |
| 1 | Ravi | 20 | Chennai |
| 2 | Kumar | 21 | Madurai |
| 3 | Anu | 19 | Coimbatore |
```

```
SQL:
SHOW GLOBAL STATUS LIKE 'Threads_connected';
```

```
After:
Global status
```

398 KILL CONNECTION

```
Before Table (students):
| id | name | age | city |
| 1 | Ravi | 20 | Chennai |
| 2 | Kumar | 21 | Madurai |
| 3 | Anu | 19 | Coimbatore |
```

```
SQL:
KILL 1;
```

```
After:
Connection killed
```

399 FLUSH TABLES

```
Before Table (students):
| id | name | age | city |
| 1 | Ravi | 20 | Chennai |
| 2 | Kumar | 21 | Madurai |
| 3 | Anu | 19 | Coimbatore |
```

```
SQL:
FLUSH TABLES;
```

```
After:
Flushed
```

400 FLUSH LOGS

```
Before Table (students):
| id | name | age | city |
| 1 | Ravi | 20 | Chennai |
| 2 | Kumar | 21 | Madurai |
| 3 | Anu | 19 | Coimbatore |
```

```
SQL:
FLUSH LOGS;
```

```
After:
Logs flushed
```